Determinants Formulas and Cofactors – Lecture #19 3/26/26

Det Properties → det $I = 1$
→ sign reverse with row exchange
→ det is linear in each row separately

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \rightarrow \overset{③}{=} \begin{bmatrix} a & 0 \\ c & d \end{bmatrix} + \begin{bmatrix} 0 & b \\ c & d \end{bmatrix}$$

① $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ ② $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ ③ $= \begin{bmatrix} a & 0 \\ c & 0 \end{bmatrix} + \begin{bmatrix} a & 0 \\ 0 & d \end{bmatrix} + \begin{bmatrix} 0 & b \\ c & 0 \end{bmatrix} + \begin{bmatrix} 0 & b \\ 0 & d \end{bmatrix}$

↓ det 0 ad $-bc$ det 0

2x2 The method of finding det using properties and principles

3x3

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} a_{11} & 0 & 0 \\ 0 & a_{22} & 0 \\ 0 & 0 & a_{33} \end{bmatrix} + \begin{bmatrix} a_{11} & 0 & 0 \\ 0 & 0 & a_{23} \\ 0 & a_{32} & 0 \end{bmatrix} + \begin{bmatrix} 0 & a_{12} & 0 \\ a_{21} & 0 & 0 \\ 0 & 0 & a_{33} \end{bmatrix} + \begin{bmatrix} 0 & a_{12} & 0 \\ 0 & 0 & a_{23} \\ a_{31} & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & a_{13} \\ a_{21} & 0 & 0 \\ 0 & a_{32} & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & a_{13} \\ 0 & a_{22} & 0 \\ a_{31} & 0 & 0 \end{bmatrix}$$

"survivor 1" "survivor 2" . . . . . .

↓ notice one component from each column

$$a_{11}a_{22}a_{33} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31}$$

Now we have a formula for the determinant that came from the 3 det properties

↓

$$\det A = \sum_{n!\text{ terms}} \pm a_{1\alpha} a_{2\beta} a_{3\gamma} \dots a_{n\omega}$$

$\alpha, \beta, \gamma \dots \omega$ = permutation of $(1, 2, \dots, n)$

Example

$$\begin{bmatrix} 0 & 0 & \perp & \top \\ 0 & \perp & \top & 0 \\ \perp & \top & 0 & 0 \\ \top & 0 & 0 & \perp \end{bmatrix}$$

eyeball the 4x4 matrix
24 possibilities from $n!$ $(4! = 24)$
22 are 0's → leaving 2 terms

$-1 + 1 = 0$

Cofactors 3x3

$\det = a_{11}(a_{22}a_{33} - a_{23}a_{32})$
$+ a_{12}$
$+ a_{13}$

3/31/26

Cofactor 3×3

$$\det = a_{11}(a_{22}a_{33} - a_{23}a_{32}) + a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(\quad)$$

$$\rightarrow \begin{bmatrix} a_{11} & 0 & 0 \\ 0 & a_{22} & a_{23} \\ 0 & a_{32} & a_{33} \end{bmatrix} + \begin{bmatrix} 0 & a_{12} & 0 \\ a_{21} & 0 & a_{23} \\ a_{31} & 0 & a_{33} \end{bmatrix} + \begin{bmatrix} 0 & 0 & a_{13} \\ - & - & 0 \\ - & - & 0 \end{bmatrix}$$

$\downarrow$

zeros out the components in same row & col of a_{11} to leave a smaller guy (2×2)

$\underbrace{\text{The cofactor of } a_{ij}}_{= C_{ij}}$: $\pm \det$ (n-1 matrix with row i col j erased)

how to determine + or − ? + if $i+j$ is even
− if $i+j$ is odd

Cofactor formula (along row 1)

$$\boxed{\det A = a_{11}C_{11} + a_{12}C_{12} + \cdots + a_{1n}C_{1n}}$$

The determinant is a products of the pivots ?

Example (real idea is to do n×n)

$$A_4 = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$[A_1] = 1 \quad [A_2] = 0 \quad [A_3] = -1$

$[A_4] = (1 \cdot [A_3]) + (-1 \cdot [A_2]) = -1 - 0 = -1$

Series

$[A_4] = -1 \sim [A_5] = 0$
$[A_6] = 1$
$[A_7] = 1$

$$[A_n] = [A_{n-1}] - [A_{n-2}]$$